

OSSP (Operating Systems And Systems Programming) - 25CS2104E SKILL -2

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX 100

char getch() {
    struct termios oldt, newt;
    char ch;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    ch = getchar();

    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    return ch;
}

int main() {
    char input[MAX];
    int index;

    printf("Simple Shell\n");
    printf("Type a command (type 'exit' to quit)\n");

    while (1) {
        index = 0;
        printf("\nshell> ");
        fflush(stdout);

        while (1) {
            char ch = getch();

            if (ch == '\n' || ch == '\r') {
                input[index] = '\0';
                printf("\n");
                break;
            }

            else if (ch == 127 || ch == 8) {
                if (index > 0) {
                    index--;
                    printf("\b \b");
                    fflush(stdout);
                }
            }
        }
    }
}
```

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        else if (index < MAX - 1) {
            input[index++] = ch;
            printf("%c", ch);
            fflush(stdout);
        }
    }

    if (strcmp(input, "exit") == 0) {
        printf("Exiting shell...\n");
        break;
    }

    if (strlen(input) == 0) {
        continue;
    }

    printf("You entered: %s\n", input);
}

return 0;
}

```

OUTPUT :

```

swaroop@SwaroopPC:~/OSLab$ nano Skill2.c
swaroop@SwaroopPC:~/OSLab$ gcc Skill2.c -o Skill2
swaroop@SwaroopPC:~/OSLab$ ./Skill2
Simple Shell
Type a command (type 'exit' to quit)

shell> hi
You entered: hi

shell> ls
You entered: ls

shell> exit
Exiting shell...
swaroop@SwaroopPC:~/OSLab$ |

```